



FLEET & SHORE PERSONNEL TRAINING

BWTS (Ballast Water Treatment System) Training

The Ballast Water Management Convention, how the ship's treatment system works, and the operating, sampling and recordkeeping duties it places on the crew.

BWM Convention

Company SMS

Port State Control

What We Will Cover

01

The BWM Convention

Why ballast water is regulated and what the Convention requires

02

BWTS Technology Types

UV treatment, electrochlorination and filtration compared

03

Operating Procedures

Commissioning, ballasting, deballasting and idle mode

04

Sampling & Compliance Testing

What a Port State Control inspection actually checks

05

Maintenance & Troubleshooting

Keeping the system running and fixing it when it doesn't

06

Recordkeeping

The Ballast Water Record Book, BWMP and certificate



SECTION 1

The BWM Convention

Why the world's fleet was required to stop discharging untreated ballast water, and what that means for this ship.

01

BACKGROUND

Why Ballast Water Is Regulated

Untreated ballast water is one of the main pathways by which invasive aquatic species are moved between regions.

- Ships take on ballast water in one port to maintain stability, trim and structural integrity, and discharge it in another, often thousands of miles away.
- That water carries organisms, larvae and pathogens from the loading port's ecosystem into the discharge port's ecosystem, where they can become invasive.
- The IMO's International Convention for the Control and Management of Ships' Ballast Water and Sediments (BWM Convention) entered into force on 8 September 2017.
- It applies to virtually all ships engaged in international voyages that carry ballast water, regardless of flag.
- Compliance is enforced through survey and certification, and verified in port through Port State Control inspection.



A Global Problem

Invasive species introduced via ballast water have caused serious ecological and economic damage worldwide — the Convention exists to break that chain at its source.

THE STANDARDS

D-1 and D-2: Two Ways to Comply

The Convention originally allowed two different standards; today virtually all ships must meet D-2.

D-1

Ballast Water Exchange Standard — replace at least 95% of ballast water volume in open sea, now largely superseded

D-2

Ballast Water Performance Standard — sets maximum concentrations of viable organisms permitted in discharged water

BWTS

A type-approved system fitted aboard to treat ballast water so its discharge meets the D-2 standard

D-1 Exchange vs. D-2 Treatment

D-2 compliance, via an approved BWTS, is now the standard expectation for the vast majority of ships.

	D-1 Exchange	D-2 Treatment (BWTS)
Method	Pump-through or flow-through exchange at sea	Filtration plus active treatment (UV, electrochlorination, etc.)
Where performed	Open sea, at least 200nm from land where practicable	Onboard, during ballasting/deballasting operations
Compliance basis	Volume exchanged (≥95%) or sequential method	Concentration of viable organisms in discharge
Current status	Largely phased out for ships required to fit a BWTS	Required by the applicable implementation schedule

APPLICABILITY

Certificates and Exemptions

Every ship subject to the Convention carries specific documentation confirming how it complies.



IBWMC

International Ballast Water Management Certificate, issued after survey confirming the ship's BWTS and procedures meet the Convention.



Ballast Water Management Plan (BWMP)

Ship-specific plan describing the system, procedures and safety precautions for managing ballast water on board.



Same Risk Area (SRA) Exchange

In limited, formally agreed sea areas, exchange may still substitute for treatment where treatment is not practicable.



Port State Control Verification

PSC officers check the certificate, record book and, where indicated, may require sampling to confirm actual compliance.



SECTION 2

BWTS Technology Types

Most systems combine a physical filtration stage with an active treatment stage — the technology used for the second stage varies by manufacturer.

02

THE COMMON PATTERN

Two Stages, One System

Almost every type-approved BWTS uses the same basic two-stage architecture.

- Stage one is mechanical filtration, typically an automatic backwashing screen filter, which removes larger organisms, sediment and debris.
- Stage two is active treatment, which neutralises or kills the smaller organisms and microorganisms that pass through the filter.
- The active treatment stage is where the technologies differ most: UV irradiation, electrochlorination, and chemical dosing are the three most common.
- Filtration alone is fitted on deballasting for some systems but is rarely sufficient on its own to meet the D-2 standard.
- System selection depends on the ship's ballast flow rate, available power, water salinity range, and space for the treatment skid.



Filter First

Removing larger particles and organisms in the filtration stage protects the active treatment stage downstream and keeps it working efficiently.

The Main BWTS Technologies

Each technology neutralises organisms by a different physical or chemical mechanism.



UV Treatment

Ultraviolet lamps damage organisms' DNA so they cannot reproduce; effective, but performance drops in turbid or high-UV-absorbing water.



Electrochlorination

Generates a low concentration of active chlorine (as sodium hypochlorite) from seawater on demand to disinfect the ballast water.



Chemical Dosing

Injects an approved biocide or disinfectant directly into the ballast line, later neutralised before discharge where required.



Mechanical Filtration

Automatic backwashing screen filters remove larger organisms and sediment as the first-stage barrier in nearly every system.

COMPARISON

Strengths and Limitations

No single technology is ideal in every water condition — this is why filtration is paired with all of them.

Technology	Key Strength	Typical Limitation
UV Treatment	No chemical residual; simple to operate	Reduced effectiveness in turbid or high-organic water
Electrochlorination	Generates its own disinfectant from seawater	Requires neutralisation; efficiency varies with salinity
Chemical Dosing	Consistent dosing independent of water quality	Chemical storage, handling and neutralisation requirements
Filtration Alone	Removes larger organisms and debris reliably	Cannot alone meet the D-2 standard for smaller organisms



SECTION 3

Operating Procedures

Correct operation, in the right sequence, is what actually delivers D-2 compliance — the equipment cannot compensate for a skipped step.

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BEFORE STARTING

Pre-Operation Checks

A short checklist before every ballasting or deballasting operation prevents most common BWTS problems.

- Confirm the BWTS is in automatic or manual mode as required, and that power, control system and alarms are functioning normally.
- Check filter differential pressure and UV lamp / electrolytic cell status readings against normal operating ranges before starting.
- Verify sufficient neutralising chemical (where fitted) and consumables such as UV lamps or filter elements are on hand.
- Confirm the ballast water record book and BWMP are available and that the planned operation matches an approved ballast water exchange or treatment sequence.
- Brief the duty engineer and Officer of the Watch on the planned operation, expected duration, and any known equipment limitations.



No Bypass Without Authorisation

Bypassing the BWTS is only ever permitted under the specific exceptions in the BWMP (e.g. safety of the ship) and must be logged and reported.

OPERATING MODES

The Four Operating Modes

The BWTS is designed to run in one of four defined modes at any given time.



Ballasting

Water is drawn in through the filter and active treatment unit before entering the ballast tanks.



Deballasting

Water is treated again on discharge (system-dependent) as it is pumped from the tanks overboard.



Commissioning / Recirculation

Used for initial setup, testing or maintenance, circulating water through the system without tank transfer.



Idle / Standby

System is stopped but ready, with periodic checks maintained per the manufacturer's manual to prevent fouling.

Running the System Correctly

Consistent, disciplined operation is what turns type-approval on paper into actual compliance in the water.



Do

- Follow the manufacturer's operating manual and the ship's BWMP sequence exactly
- Log every ballasting/deballasting operation, including any alarms or deviations
- Monitor UV intensity, TRO (Total Residual Oxidant) or equivalent parameters throughout
- Report a BWTS fault to the Chief Engineer and Master immediately, not after the operation



Don't

- Don't start a ballast operation with a known, unresolved system alarm
- Don't exceed the system's rated flow rate to speed up the operation
- Don't discharge ballast water without treatment except under a permitted exception
- Don't assume a previous successful cycle guarantees this one will pass without checks



SECTION 4

Sampling & Compliance Testing

04

What actually gets checked when Port State Control comes aboard, and how the crew demonstrates compliance.

What Port State Control Checks

PSC inspections use a risk-based, multi-step approach before any sample is actually taken.

- Initial inspection reviews the IBWMC, Ballast Water Record Book entries, and confirms the fitted BWTS matches the certificate.
- Inspectors assess whether operation and maintenance records are consistent, complete and correctly signed off.
- Indicative analysis — a rapid onboard or portside check of parameters such as organism viability — may follow if the initial review raises concerns.
- Detailed sampling and full compliance testing is reserved for cases where indicative analysis or documentation suggests a genuine problem.
- A ship found non-compliant can face detention, a warning, or a requirement to rectify the deficiency before departure, depending on severity.



Records Are the First Line

Complete, accurate, contemporaneous records are usually what convinces an inspector a ship is complying — gaps or inconsistencies invite closer scrutiny.

Tools Used for Sampling and Checks

Crew-operated indicative testing helps confirm the system is performing before a port inspection ever begins.



TRO Analyser

Measures Total Residual Oxidant in electrochlorination systems to confirm dosing and neutralisation are within range.



Sampling Ports

Dedicated points on the ballast line, fitted per the system design, used to draw representative water samples.



Particle / Organism Counters

Portable instruments used in some indicative analysis protocols to estimate viable organism concentration.



System Self-Monitoring Data

UV intensity, flow rate and filter differential pressure logs generated automatically by the BWTS control system.

POSSIBLE OUTCOMES

PSC Deficiency Categories

The response required depends on how serious the deficiency is judged to be.

Finding	Example	Typical Outcome
Documentation only	Missing signature or minor record book gap	Noted deficiency; rectify before next port
Operational deficiency	BWTS not operated per approved procedure	Deficiency with rectification deadline
Equipment malfunction	System inoperative, no valid contingency measure	Detention until repaired or exception applies
Non-compliant discharge	Sampling confirms discharge exceeds D-2 limits	Detention; formal investigation and reporting



SECTION 5

Maintenance & Troubleshooting

A BWTS that isn't maintained will eventually fail exactly when it's needed most — usually mid-ballasting, in port.

05

ROUTINE CARE

Keeping the System Reliable

Most BWTS failures trace back to a deferred routine maintenance task, not a sudden equipment defect.

- Clean or replace UV lamp sleeves and check lamp output regularly — fouled or aged lamps are a leading cause of reduced UV dose.
- Inspect and clean electrode cells in electrochlorination systems, as scale buildup reduces chlorine generation efficiency over time.
- Backwash and inspect filter screens per the manufacturer's schedule to prevent differential pressure alarms and reduced flow.
- Calibrate sensors — UV intensity, flow, TRO, turbidity — at the intervals specified in the maintenance manual, not only when readings look wrong.
- Keep critical spares (lamps, seals, filter elements) on the ship's spares list and reorder before stock runs low.



Trend the Data

Reviewing UV intensity or TRO trends over weeks, not just the current reading, catches slow degradation before it becomes an alarm or a failed sample.

Troubleshooting the BWTS

A structured response to a fault protects both the equipment and the ship's compliance record.



Do

- Check the alarm log and manufacturer's fault-finding table before opening any equipment
- Isolate and lock out power before inspecting UV units or electrolytic cells
- Record the fault, cause and corrective action taken in the maintenance log
- Notify the Chief Engineer and, where operation is affected, the Master



Don't

- Don't reset an alarm repeatedly without investigating its root cause
- Don't substitute non-approved spares or consumables to keep the system running
- Don't continue ballasting on a system showing a persistent treatment alarm
- Don't delay reporting a fault that could affect the next port's compliance

COMMON FAULTS

Four Fault Categories to Know

Most BWTS troubleshooting calls fall into one of these categories.

UV

Low UV intensity — fouled sleeve, aged lamp, or turbid feed water

ΔP

High filter differential pressure — clogged screen or failed backwash cycle

TRO

Out-of-range TRO — electrode fouling, low salinity, or dosing pump fault

SNS

Sensor fault — miscalibration or a wiring/connection fault giving a false alarm



SECTION 6

Recordkeeping

The paperwork is not separate from compliance — it is the primary evidence that compliance actually happened.

06

THE CORE DOCUMENTS

What Must Be Recorded

Three documents together form the ship's ballast water compliance file.

- The Ballast Water Record Book (BWRB) logs every ballasting, deballasting, treatment and exchange operation, with dates, positions and volumes.
- Entries must be made as soon as possible after each operation, signed by the responsible officer, and retained for the period set by the flag state.
- The Ballast Water Management Plan (BWMP) describes ship-specific procedures, safety precautions and the actions to take if the BWTS fails.
- The International Ballast Water Management Certificate (IBWMC) confirms survey and compliance status, and must be kept current and on board.
- Any bypass, malfunction or non-compliant discharge must be recorded with the reason, together with the corrective action taken.



Write It As It Happens

A record book completed from memory at the end of a voyage is far less credible — and less accurate — than one completed operation by operation.

What Goes Into Each BWRB Entry

A complete entry answers what was done, where, how much, and by whom.

Field	Example Entry	Why It Matters
Operation type	Ballasting / Deballasting / Exchange / Treatment	Establishes what compliance route applies
Date, time, position	Lat/long or port, UTC time	Confirms where the operation occurred
Tank(s) and volume	No. 2 WBT, 850 m ³	Links the operation to specific ballast spaces
Signature	Officer of the Watch / Chief Engineer	Establishes accountability for the entry

WHO IS RESPONSIBLE

Recordkeeping Responsibilities

Ballast water recordkeeping is a shared duty across the ship's senior officers.



Master

Holds overall responsibility for the ship's compliance with the BWM Convention and its BWMP.



Chief Engineer

Responsible for BWTS operation, maintenance and the technical entries in the record book.



Chief Officer

Coordinates ballasting/deballasting with cargo and stability planning, and confirms tank-level entries.



DPA / Company

Reviews records during audits and supports the ship in resolving equipment or compliance issues.



COURSE COMPLETE

A well-run BWTS protects the marine environment and keeps the ship moving — through every port, every time.

Company DPA — 24/7 emergency line

Fleet Technical Department — mng@gloryship.com.tr

Fleet Safety Department — mng@gloryship.com.tr